

PENG YIFENG

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EDUCATION

CUHK-Shenzhen , Finance (Financial Technology), <i>MSc</i>	2021.09 - 2023.07
The University of Hong Kong , Computer Science, <i>MSc</i>	2020.09 - 2021.09
Hunan University , Computer Science and Technology, <i>B.Eng.</i>	2016.09 - 2020.06

SELECTED RESEARCH (LATEST RESEARCH PROPOSAL)

Hidden Memorization Trap in Financial LLMs 2025.11 - Present

- **Collaborators:** XIE Lab of AI4Sci (Wuhan University / Yale University / Dartmouth College).
- **Abstract:** This study questions pseudo-forecasting in financial LLMs: models may not infer historical outcomes from evidence, but instead recall macro indicators, prices, news and earnings-call facts already inside their training window.
 - Led deanonymization experiments on masked earnings-call transcripts, measuring how strongly mainstream models can recover entity information from anonymized financial text.
 - Tested artificial/fake knowledge-cutoff prompts and entity-neutering safeguards, evaluating the limits of mainstream anti look-ahead-bias controls in financial settings.
 - Helped organize evidence across macro indicators, market indices, portfolio prices and earnings calls, arguing that LLM-based finance research should prioritize post-cutoff or temporally consistent evaluation.

Poison Pills: Insights from Vulnerability Disparity Among LLMs 2024.05 - 2025.06

- **Collaborators:** Chen Chen and Wu Zhizheng (CUHK-Shenzhen)
- **Abstract:** This study proposes the poison-pill attack: using a very small set of near-duplicate, locally mutated factual examples (time, location, entity, etc.) to contaminate targeted knowledge while keeping overall benchmark performance largely unchanged.
 - Independently completed the main experiments: constructed dominant/long-tail topic pairs, generated poison-pill data, deployed and fine-tuned open-source LLMs, ran repeated-query evaluation and compared random/multi-position contamination baselines.
 - Systematically evaluated clean-corpus dilution, model scale, pruned/distilled/compressed models, collateral damage and simultaneous attacks; found that long-tail knowledge and compressed models are more vulnerable, while dominant knowledge can spread contamination through associative memory.
 - Took primary responsibility for manuscript writing and result presentation, including motivation, method definition, experimental design, result analysis and security implications.

Internet public opinion, investor sentiment and excessive trading 2022.08 - 2023.05

- **Independent project:** Based on 2018–2022 A-share market samples, scraped Eastmoney Guba text and matched it with Wind intraday trading data to construct hourly investor-sentiment indices and overtrading measures.
 - Built and manually validated a Chinese financial sentiment dictionary, then used hourly panel regressions to study the effect of investor sentiment on intraday overtrading.
 - Examined heterogeneity across institutional/retail investors, bull/bear markets and large/mid/small-cap stocks; found that positive sentiment significantly amplifies intraday overtrading, with effects varying by investor type and market environment.

PUBLICATIONS & WORKING PAPERS (GOOGLE SCHOLAR)

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- **[AACL-IJCNLP]** Peng, Y., Wu, Z., & Chen, C. (2025). Swallowing the Poison Pills: Insights from Vulnerability Disparity Among LLMs.
 - **[Working On]** Xie, Q., Lin, Q., Duan, J., & Peng, Y. Forecasting or Flashback? The Hidden Memorization Trap in Financial LLMs
 - **[Revising]** Zhang, Y., Gao, C., & Peng, Y. (2025). CDiff: Bridging Conditional Diffusion and Causal Inference with Statistical Guarantees via Principled Distribution Approximation
 - **[FMCG]** Peng, Y. & Gao, C. (2025). Convenience at a Cost: The Paradox of Internet-Based Fund Distribution Channels and Investment Returns

- [\[Preprint\]](#) Peng, Y. (2024). High-Frequency Investor Sentiment on Intraday Overtrading
- [\[Book Chapter\]](#) Chen, C., Peng, Y., Hong, J., & Lu, Q. (2023). Future development of digital built asset management. *Digital Built Asset Management*, 304–340.

RESEARCH & PROFESSIONAL EXPERIENCE

CUHK-Shenzhen & Wuhan University, Research Associate 2023.07 - 2026.05

- Worked with Prof. Chen Chen and Prof. Qianqian Xie's team on financial-LLM safety, human-AI alignment in high-risk finance, look-ahead bias control and reliability of domain-specific LLMs.
- Built and executed LLM experiment pipelines, including poisoned-corpus construction, deployment/fine-tuning, repeated evaluation, anonymization/deanonymization tests, statistical analysis and visualization.
- Contributed to literature reviews, research design, manuscript writing and submission materials while coordinating progress across cross-institutional collaborators.
- Assisted undergraduate teaching, course preparation and weekly academic seminar organization.

ZEEKR Intelligent Technology, Operations Intern 2024.03 - 2024.09

- Quantitatively evaluated how offline pre-sales service quality relates to conversion from refundable deposits to firm orders.
- Supported digital transformation in operations by structuring metrics for pre-sales consultation, outbound calls, lead circulation and deposit-to-order conversion.
- Organized store-level operating and conversion data to support pre-sales resource allocation and workflow optimization.

ACTIVITIES & SERVICE

Shaw College, CUHK-Shenzhen, Peer Tutor (D-Block) 2022.05 - 2024.05

- Mentored students on floors 10–11 of D-Block through regular dorm visits, mental-wellbeing check-ins, academic consultation and mentoring talks; received the [Shaw Spirit Award](#).

CUHK-Shenzhen Shimmer Charity Organization, Long-Term Volunteer 2021.09 - 2025.05

- Maintained order at COVID-19 testing sites and provided long-term support for elderly-care visits and charity activities at Longgang Welfare Center.
- Received the [Outstanding Volunteer](#) honor.

HONORS, COMPETITIONS & REVIEWING

- **Academic service:** ICML 2026 [Gold Reviewer](#); NeurIPS 2026 Reviewer.
- **Kaggle:** Diabetes Prediction Challenge [Top 10%](#) (167/4,168, 2025); Forecasting Sticker Sales [Top 5%](#) (25/2,722, 2025); Jane Street Real-Time Market Data Forecasting [Top 10%](#) (119/2,757, 2024).
- **Scholarships & honors:** Research Commendation Award (2024); Shenzhen Finance Institute Scholarship (2024); Hunan University First-Class President Scholarship (2017–2019).
- **Competitions:** 13th Challenge Cup Provincial Special Prize (2019); Hunan University Information and Electronic Engineering Competition First Prize (2018).